Assignment: Movie Review Data Analysis

Student Details

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Introduction

In this assignment, I have worked with a real-world movie review dataset, performing text data analysis using Python libraries such as Pandas and Numpy. I formulated 20 problem statements based on the dataset and applied appropriate methods to find the solutions. Screenshots of the code execution are attached below each problem statement.

NOTE : This is common code before running all 20 code

```python
# ------------------------------------------
# 📦 Importing Required Libraries
# ------------------------------------------
import pandas as pd
import numpy as np
import os
import re
from collections import Counter

# ------------------------------------------
# 📁 Load the IMDB Review Dataset
# ✅ Replace this with your own path
# ------------------------------------------
file_path = '/content/IMDB Dataset.csv'  # <-- Change only this line!
df = pd.read_csv(file_path, encoding='utf-8', on_bad_lines='skip')

# ------------------------------------------
# 🧼 Basic Data Preprocessing
# ------------------------------------------

# 1. Strip HTML tags from reviews (optional but useful)
df['review'] = df['review'].str.replace(r'<.*?>', '', regex=True)

# 2. Add word count column
df['word_count'] = df['review'].apply(lambda x: len(str(x).split()))

# 3. Optional: Convert all reviews to lowercase for consistency
df['review'] = df['review'].str.lower()

# ✅ Preview first few rows
print(df.head())
```

Problem Statements and Solutions

Problem 1

Problem Statement: Display the total number of reviews in the dataset.

Solution:

```
[3] print('Total reviews:', len(df))
```

```
Total reviews: 50000
```

Problem 2

Problem Statement: Count how many reviews are labeled as 'positive' and 'negative'.

Solution:

```
[4] print(df['sentiment'].value_counts())
```

```
sentiment
positive    25000
negative    25000
Name: count, dtype: int64
```

Problem 3

Problem Statement: Find the average length of the reviews (in characters).

Solution:

```
[5] print('Average review length:', df['review'].apply(len).mean())
```

```
Average review length: 1285.17476
```

Problem 4

Problem Statement: Add a new column showing the word count of each review..

Solution:

```
[6] df['word_count'] = df['review'].apply(lambda x: len(str(x).split()))
    df[['review', 'word_count']].head()
```

	review	word_count
0	one of the other reviewers has mentioned that ...	301
1	a wonderful little production. the filming tec...	156
2	i thought this was a wonderful way to spend ti...	162
3	basically there's a family where a little boy ...	132
4	petter mattei's "love in the time of money" is...	222

Problem 5

Problem Statement: Find the review with the maximum word count.

Solution:

```
[7] max_review = df.loc[df['word_count'].idxmax()]
    print(max_review['review'])
```

```
match 1: tag team table match bubba ray and spike dudley vs eddie guerrero and chris benoit bubba ray and spike dudley started things off with a tag team table match against eddie guer
```

Problem 6

Problem Statement:Check if there are any missing values in the dataset..

Solution:

```
[8] print(df.isnull().sum())
```

```
review        0
sentiment     0
word_count    0
dtype: int64
```

Problem 7

Problem Statement: Calculate the average word count for positive reviews.

Solution:

```
[9] print(df[df['sentiment'] == 'positive']['word_count'].mean())
```

```
228.92608
```

Problem 8

Problem Statement: Calculate the average word count for negative reviews.**Solution:**

```
[10] print(df[df['sentiment'] == 'negative']['word_count'].mean())
```

```
225.29808
```

Problem 9

Problem Statement: Count how many reviews contain the word 'great'.

Solution:

```
[11] print(df['review'].str.contains('great', case=False).sum())
```

```
13807
```

Problem 10

Problem StatementShow the first 5 positive reviews that mention 'bad'.

Solution:

```
[12] df[(df['sentiment'] == 'positive') & (df['review'].str.contains('bad', case=False))].head()['review']
```

	review
14	this a fantastic movie of three prisoners who ...
16	some films just simply should not be remade. t...
20	after the success of die hard and it's sequels...
26	"the cell" is an exotic masterpiece, a dizzyin...
45	as a disclaimer, i've seen the movie 5-6 times...

dtype: object

Problem 11

Problem Statement: Remove HTML tags from reviews using regex..

Solution:

```
[13] df[(df['sentiment'] == 'positive') & (df['review'].str.contains('bad', case=False))].head()['review']
```

	review
14	this a fantastic movie of three prisoners who ...
16	some films just simply should not be remade. t...
20	after the success of die hard and it's sequels...
26	"the cell" is an exotic masterpiece, a dizzyin...
45	as a disclaimer, i've seen the movie 5-6 times...

dtype: object

Problem 12

Problem Statement: Count the number of unique reviews in the dataset.

Solution:

```
[14] print('Unique reviews:', df['review'].nunique())
```

```
Unique reviews: 49581
```

Problem 13

Problem StatementDisplay the distribution (min, max, mean) of word counts.

Solution:

```
[15] print(df['word_count'].describe())

count    50000.000000
mean       227.112080
std        168.275619
min          4.000000
25%        124.000000
50%        170.000000
75%        275.000000
max       2450.000000
Name: word_count, dtype: float64
```

Problem 14

Problem Statement: Show the 10 most frequent words in all reviews.

Solution:

```
[16] from collections import Counter
     all_words = ' '.join(df['review']).lower().split()
     common_words = Counter(all_words).most_common(10)
     print(common_words)

[('the', 640480), ('a', 316783), ('and', 313811), ('of', 286693), ('to', 264638), ('is', 204891), ('in', 180040), ('i', 142472), ('this'
```

```
[15] print(df['word_count'].describe())
```

Problem 15

Problem Statement: Find reviews that mention both 'love' and 'hate'.

Solution:

```
[17] df[df['review'].str.contains('love', case=False) & df['review'].str.contains('hate', case=False)]['review'].head()
```

	review
75	it tries to be the epic adventure of the centu...
210	i have just given a 10 for thieves highway, i ...
218	this movie really woke me up, like it wakes up...
277	some comments here on imdb have likened dog bi...
278	i watched this movie for the hot guy--and even...

dtype: object

Problem 16

Problem StatementCalculate the proportion of reviews with more than 100 words.

Solution:

```
[18] prop = (df['word_count'] > 100).mean()
     print(f'Proportion: {prop:.2%}')
```

```
Proportion: 86.98%
```

Problem 17

Problem Statement: Get the average number of characters in positive reviews.

Solution:

```
[19] print(df[df['sentiment'] == 'positive']['review'].apply(len).mean())
```

```
1301.27336
```

Problem 18

Problem Statement: Convert sentiment labels to numeric (positive = 1, negative = 0).

Solution:

```
[20] df['sentiment_numeric'] = df['sentiment'].map({'positive': 1, 'negative': 0})
     df[['sentiment', 'sentiment_numeric']].head()
```

	sentiment	sentiment_numeric
0	positive	1
1	positive	1
2	positive	1
3	negative	0
4	positive	1

Problem 19

Problem Statement: Create a boolean column has_excellent if 'excellent' appears in the review.

Solution:

```
[21] df['has_excellent'] = df['review'].str.contains('excellent', case=False)
     df[['review', 'has_excellent']].head()
```

	review	has_excellent
0	one of the other reviewers has mentioned that ...	False
1	a wonderful little production. the filming tec...	False
2	i thought this was a wonderful way to spend ti...	False
3	basically there's a family where a little boy ...	False
4	petter mattei's "love in the time of money" is...	False

Problem 20

Problem Statement: Find how many reviews have both more than 150 words and are labeled positive.

Solution:

```
[22] df[(df['word_count'] > 150) & (df['sentiment'] == 'positive')].shape[0]

14581
```